Benjamin Daniel Poole

benjamindpoole@protonmail.com | benjamindpoole.com | linkedin.com/in/benjamin-d-poole | github.com/Bpoole908

Education

University of North Carolina at Charlotte, *PhD in Computer Science, GPA 4.0* 2019-Expected 2026
University of North Carolina at Charlotte, *MS in Computer Science, GPA 4.0* 2017–2019
University of North Carolina at Charlotte, *BS in Computer Science, GPA 3.9* 2014–2018

Technical Skills

Languages: Python, Java, Go, C#
Expertise: Machine Learning, Deep Learning, Reinforcement Learning (RL), RLHF, Graph Learning, Linear Algebra, Statistics
Tools: PyTorch, TensorFlow, Keras, Sklearn, NumPy, Git, Docker, Linux, Slurm

Experience

Research Assistant, CharMLab – Charlotte, NC 2019-Present

Feedback Manipulation Regularization for Offline Alignment

- Led team developing model-agnostic algorithm leveraging human feedback to improve offline imitation learning alignment by 24–95% over baselines.
- Designed and implemented pipelines for collecting human demonstrations and feedback, and performing training and evaluation of popular online/offline RL algorithms.

Value-Based Data Rehearsal for continual RL

- Led team investigating value-function regularization multi-cyclic continual RL settings.
- Developed strategies that, on average, improve performance up to 20% while reducing catastrophic forgetting and increasing between task transfer by 41%.

Data-Driven Graph Construction (github.com/RL-BCI-Lab/gc4eptn)

- Co-lead for project applying Gaussian graphical models to construct graphs of electrical power transmission networks from real-time simulated time series data.
- Achieved 98% graph recovery on flow graphs and 93% network graph reconstruction across all topologies and load conditions.
- Collaborated with domain specialists to gather real-time simulated time series data.

Error-related Potential Variability (github.com/RL-BCI-Lab/deepbci)

- Led team investigating transferability of ML models (CNN/SVM) between error-relate potential brain signal variations.
- Designed and conducted 10+ human subject BCI experiments using consumer-grade EEG equipment.
- Implemented comprehensive pipelines for preprocessing, visualization, training, and evaluation.

Primary Instructor and Co-Instructor, UNC Charlotte – Charlotte, NC 2019-2025

- Served as primary instructor for two semesters of intro ML and co-instructor for 10+ semesters across intro and applied ML courses with 50+ students each.
- Developed curriculum for intro ML, applied ML, and mathematics for ML courses.

Data Analytics Intern, Klarrio – Apex, NC May-Aug 2019

- Helped research and prototype stateful streaming prototype using Pulsar for visualizing Twitter analytics data using AWS, Pulsar, Docker, Python, and Java.
- Demonstrated the usability of new streaming technology for the company to integrate into their workflow.

Software Engineer Intern, IBM – Durham, NC May-Aug 2017

- Worked alongside of developers and users to develop quality of life CLI tools for IBM's cloud open source FaaS project OpenWhisk.
- Implemented CLI tools in Go and developed production level test cases in Scala.

Publications

Data-Driven Graph Construction of Power Flow Graphs for Electric Power Transmission Networks. ICMLA, 2024. 10.1109/ICMLA61862.2024.00053

Towards interactive reinforcement learning with intrinsic feedback. Neurocomputing, 2024. 10.1016/j.neucom.2024.127628

Error-related Potential Variability: Exploring the Effects on Classification and Transferability, SSCI, 2022. 10.1109/SSCI51031.2022.10022137